

Task 6

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C.

Does the low-frequency signal contain mostly bass and the high-frequency signal contain mostly cymbals and "sizzle"?

Low-Frequency Signal (Woofer Output) — [output_low_freq.wav](#)

- Contains mostly bass, deep tones, drums, and background harmony.
- Muffled overall — lacks clarity in vocals, cymbals, or strings.
- Emphasizes frequencies below 2 kHz, which is where most of the body of music lives.

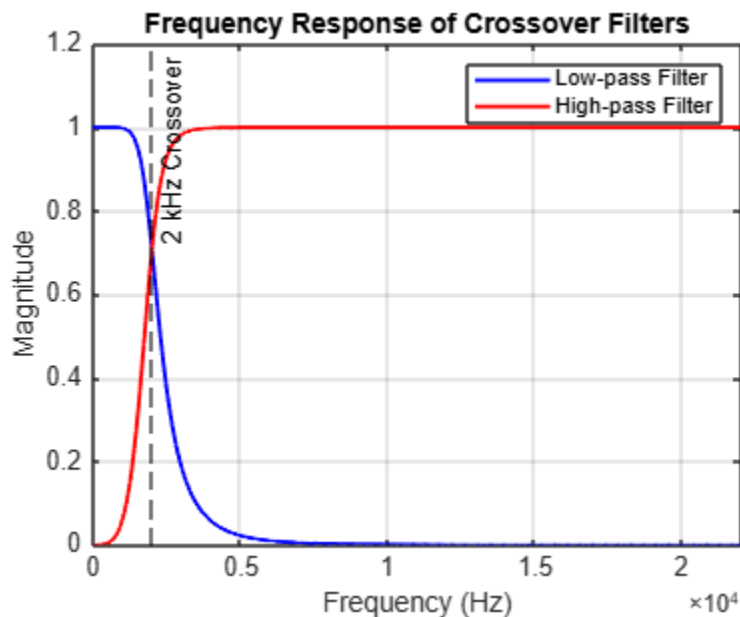
High-Frequency Signal (Tweeter Output) — [output_high_freq.wav](#)

- Contains mostly sharp, crisp elements, like cymbals, hi-hats, violins, or breath sounds.
- Very thin and airy — sounds lack fullness or bass presence.
- Focuses on frequencies above 2 kHz, contributing to detail and brightness.

The low-frequency signal contains mostly bass, and the high-frequency signal contains the sizzle and sparkle like cymbals and high harmonics.

Plot the frequency response of both filters on the same graph to visualize the crossover point.

This is the plot output:



This plot shows the frequency response of a low-pass (blue) and high-pass (red) Butterworth filter with crossover at 2 kHz. The two filters intersect at this point, both having roughly 0.707 (-3 dB) magnitude, as the Butterworth filters would suggest. This offers an even combined response at the crossover.

The low-pass filter lets frequencies lower than 2 kHz pass with minimal attenuation and rolls off without ripple after the cutoff frequency.

The high-pass filter reverses this: it rolls off without ripple prior to the cutoff frequency and lets frequencies above it pass. Roll-off is steep, as in a 4th-order Butterworth filter (~24 dB/octave), and the transition is smooth and monotonic without ripple.

The response confirms correct filter design, with flat passbands and high stopband attenuation. The crossover is well situated at 2 kHz, with a natural transition between frequency bands.